

“It’s Teamwork!” – Cooperation and Collaboration in Social Casual Exergames

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Abstract

Exergames offer an engaging approach to combat sedentary lifestyles and promote physical activity. While the benefits of multiplayer gaming are established, the nuanced effects of different cooperative modes remain underexplored for exergames. This study investigates the impact of cooperation (parallel work towards a shared goal) versus collaboration (joint effort towards a shared goal) on player experience in casual exergames. Results revealed that collaboration significantly improved both positive and negative affect, fostered stronger interpersonal connections, and was preferred by most players, while cooperation often was perceived as competitive and led to higher exertion levels. These findings illuminate the complex interplay between cooperation, competition, and collaboration in exergames.

CCS Concepts

• **Applied computing**; • **Computers in other domains**; • **Personal Computers and PC applications**; • **Computer games**;

Keywords

Casual exergames, Cooperation, Collaboration, Player experience

ACM Reference Format:

Helena Orädd, Erik Kristoffer Berglund, and Aseel Berglund. 2025. “It’s Teamwork!” – Cooperation and Collaboration in Social Casual Exergames. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA ’25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3706599.3720041>

1 Introduction

Prolonged sedentary behavior is associated with numerous health risks and increased mortality rates [1, 2] independent of the duration of moderate-to-vigorous physical activity [1]. Physical activity is essential for everyone and regularly breaking up prolonged periods of sitting offers health benefits, regardless of overall sedentary time [3]. The health recommendations are a minimum of 150 minutes a week of moderate and vigorous physical activity [4] and breaking up sitting time every 20-30 minutes with 2-3 minutes of

light activity [5, 6]. Exergaming combines gameplay with physical exercise [7] and has shown promise in increasing physical activity [8-10] and reducing sedentary behaviors [11-13]. Exergames serve two main purposes; delivering physical exercise benefits and providing enjoyable gaming experiences. Exergames can also foster positive social relations through play and playfulness [14] and increase positive affect in players [15, 16].

Social interaction with others is a strong motivator for continued exergame play [17] and leads to higher adherence to exergaming [18]. Self-determination theory (SDT) is often used to explain the motivational effect of social interaction in games [17]. According to SDT, when people’s three basic needs are met, they become intrinsically motivated to engage in an activity [19]. Those needs are competence (increasing skill), autonomy (being in control of decisions), and relatedness (developing and maintaining relationships) [19], all of which have been shown to be related to higher enjoyment [20]. Working together in a cooperative or collaborative setting can increase social interaction between players [21, 22], and has been found to increase motivation, adherence, and self-efficacy to a higher degree than competition [22]. While competition is usually found to be more effective for increasing exertion [22], in some situations working together can be more effective in increasing both motivation and effort [23]. However, little to no research has been done on how different ways of working together can affect the experience of exergaming. In this paper, we will examine the research question: *How is the experience of playing an exergame affected by playing in a cooperative or a collaborative setting?*

2 Related Work

In everyday speech, the words “cooperative” and “collaborative” are used interchangeably to describe any situation in which multiple people work together to reach a goal. However, multiple fields make a distinction between the two words. Some fields define cooperation as a loosely coupled teamwork in which a problem can be split into separate parts to work in parallel towards a common goal, while collaboration is more tightly coupled with all participants solving the problem jointly in a coordinated effort [24, 25]. Likewise, works viewing different types of teamwork through a lens of power dynamics classifies cooperation as having a “power to” structure while collaboration has a “power with” structure [26]. The power-to structure allows higher independence as well as fragmentation of work, giving members higher autonomy over their own work. The power-with structure builds stronger interpersonal relationships through a strong focus on the collective where nobody holds power over other members. While these definitions follow the common

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CHI EA ’25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/2025/04

<https://doi.org/10.1145/3706599.3720041>

theme of cooperation being more individualistic while collaboration requires members to work closely together, this is not a universal definition. In game research specifically there does not seem to exist a common definition, with different researchers using nearly opposite definitions from each other. For example, while Baek and Touati [27] also describe cooperation as dividing a common task into individual subtasks while collaboration requires simultaneous work on the same task, Beznosyk et al. [28] describe a cooperative virtual environment as one where multiple individuals must interact with an object at the same time to solve a task while collaboration only allows one individual to handle an object at a time.

2.1 Cooperation and Collaboration in Games

Many game studies make no distinction between collaboration or cooperation, instead focusing on social interdependency. Social interdependence is when multiple individuals share some common goals, each individual's outcome is affected by the actions of others [29]. Interdependence exists in both cooperation, where the common goal is a shared goal by all participants, and in competition, where the common goal is one that only one or a few individuals can achieve [30]. Interdependence can be positive, which typically leads interactions where people assist and encourage each other, or negative, where people typically try to obstruct or discourage each other's efforts [30]. Cooperation and interdependency can then be viewed as two separate concepts. Depping and Mandryk [31] define cooperation as having a shared goal (in contrast to competition with opposing goals) while interdependency refers to the players' ability to directly affect the other player with their actions (in contrast to independency players' actions do not affect the other's). They analyzed gameplay of pairs of strangers playing a puzzle game with four variations (independent cooperation, interdependent cooperation, independent competition, and interdependent competition) and found that both cooperation and interdependence individually led to higher enjoyment and perceived connectedness. Additionally, cooperation led to higher effort in the game than competition regardless of dependency level, but if the cooperation was interdependent, it could also increase tension in players. Harris and Hancock [32] explored the effect of interdependency on player experience in a two-player cooperative game designed with three levels of interdependency: low (e.g., players can take independent actions), medium (e.g., players must do an action each in sequence), and high (e.g., players must take an action each in quick succession to each other). They found that higher interdependence led to stronger feelings of social connectedness and higher interest, effort, perceived importance, and engagement. However, they also noted that high interdependency can become tedious if players for example must take turns waiting for each other's actions before being able to continue playing.

In this study, we distinguish between cooperation and collaboration in the context of cooperative exergames by their varying levels of interdependency, using the following terminology throughout the rest of the paper:

- **Cooperation:** Working towards a common goal while maintaining individualism and allowing division of work tasks. Low social interdependency between individuals' actions.

- **Collaboration:** Working together in a joint effort to reach a common goal that could not be achieved by an individual. High social interdependency between individuals' actions.

2.2 Cooperative Exergames

Many studies have been done to compare competitive and cooperative exergaming. However, many of those studies still had competition included in their cooperative game modes. For example, when investigating how competitive and cooperative exergames affect anxiety in male university students, Boyang et al. [33] used a mobile exergame where players competed one-on-one or in teams of two to complete the most push-ups in one minute. When playing in a team, the players' pushups were summed up for the pair's final score. In another study Staiano et al. [34] split participants into a competitive group where players competed against each other one-on-one by expending calories and earning points, and a cooperative group where participants worked together in pairs to earn a shared final score. While the cooperative group did not compete directly with other pairs, both groups had their weekly leaderboard where the names of the highest performing individuals or team were displayed at the end of each week, leading to a sense of competition to score high. While still having an aspect of competition, both studies showed positive effects from using cooperative play modes; when compared to pure competition, cooperation led to higher intrinsic motivation and enjoyment of the game design [34] and significantly improved state anxiety and anxiety-related physiological behaviors when competition did not [33].

Other studies have instead compared competing with the computer or another human and playing as a team against the computer. Kim et al. [35] compared single-player competition against the computer, multi-player one-on-one competition, and cooperation in pairs against the computer, finding higher activity and heart rate in the competitive version compared to both the cooperative and single-player variants. Participants also preferred playing with or against a human partner over a virtual one, showing the importance of exergames that can be played with other people. However, this preference is not necessarily true for everyone or in any situation. When studying competitive and cooperative attitudes, Chan et al. [36] found that competitive individuals tend to prefer to play alone while cooperative individuals enjoy playing together with other people. In addition, pairs of friends enjoyed playing together more than pairs of strangers.

2.3 Collaborative Exergames

In a study by Qiu et al. [37], a collaborative screen-based exergame was used to explore how virtual and co-located gameplay affected the experience of social exergames. For both settings, each player controlled an avatar by tilting their body on a balancing board. The two avatars were connected via an elastic band forming an area that could be stretched and squished together as the players moved their avatars on the screen. To gain points, players had to move their avatars to stretch the area to touch some objects while avoiding others. The study showed that players felt more connected with their co-player when playing the collaborative versions of the exergame, both virtual and co-located, than when playing a version

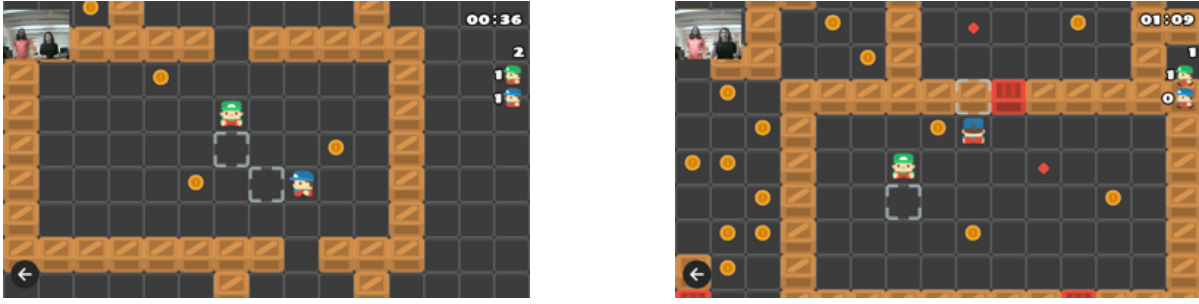


Figure 1: Cooperative mode (left) and collaborative mode (right) of the exergame Coin Run.

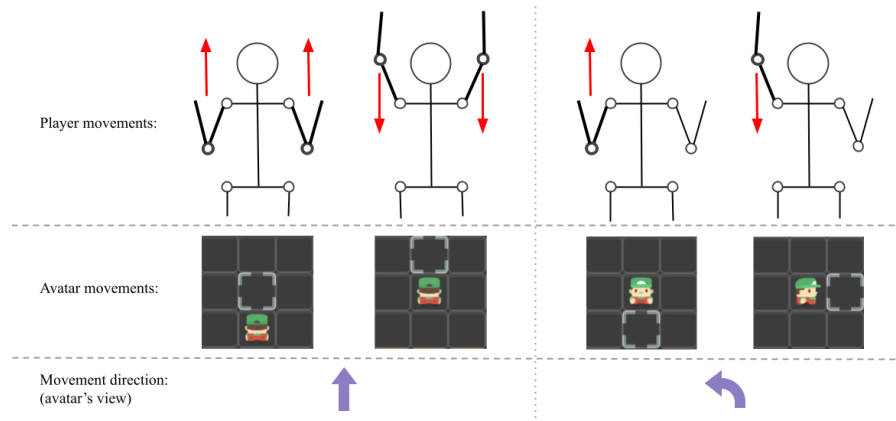


Figure 2: Movements used in the exergame Coin Run. Moving forward (left) and rotating (right).

where players were only aware of the other player's avatar as they played non-collaboratively in parallel.

Hwang et al. [38] explored how gameplay in a collaborative exergame was affected by different learning styles by setting up three versions of the exergame: one using only visual cues to communicate with a teammate, one allowing communication through speech but no visual cues, and one using both visual and auditory communication. With visual cues, the players could perform movements to guide their teammate or synch movements. Experimental results showed improved collaboration when using both visual cues and verbal communication, but inconsistent player behavior when using only auditory or visual cues, meaning the task performance was not improved.

3 Methods

Coin Run is a co-located two-player exergame developed in Godot. The game is played in front of a single computer screen using a web camera and TensorFlow to track players' positions and poses. Each player controls their own avatar, navigating through a series of interconnected, walled-off rooms. The objective is to collect as many coins as possible in two minutes, with the game's final score being the combined number of coins. This scoring system encourages teamwork while still allowing for a sense of personal achievement. To support our research objectives, two distinct game mechanics

were implemented: A cooperative mode where all passages between rooms remain open to allow independent movements, and a collaborative mode where one player must activate a trigger to let the other pass through the doors (see Figure 1).

The exergame is controlled through arm movements. Lifting one arm rotates the avatar in that direction, while lifting both arms move it forward one step (Figure 2) if the path is free. Players can see themselves on screen via the webcam feed to ensure proper positioning for motion tracking. By providing immediate visual feedback, players can optimize their stance and movements to guarantee accurate motion capture throughout the gameplay experience.

3.1 Measures

Physical intensity was assessed through subjective ratings using the Borg Rating of Perceived Exertion (RPE) scale [39, 40] (ranging from 6-20) and objective heart rate monitoring via Polar Unite fitness trackers, where exercise intensity percentages were calculated by comparing the heart rates from gameplay sessions to estimated maximum heart rates (220 minus age). Psychological measurements included mental state (using the 10-item Positive and Negative Affect Scales, PANAS, [41]), need satisfaction (using the Player Experience of Need Satisfaction, PENS, questionnaire [19]), physical activity enjoyment (using the Physical Activity Enjoyment Scale, PACES [42, 43]), and achievement through game scores.

3.2 Participants and Procedure

The study involved 48 participants with ages between 19 and 50 ($M = 26.56$, $SD = 7.04$) years old. The gender distribution was 43.8% ($n = 21$) men, 39.6% ($n = 19$) women, 14.6% ($n = 7$) non-binary, and 2.1% ($n = 1$) unsure of gender identity. When rating their familiarity with exergames on a scale from 1 (“I have never played an exergame”) to 5 (“I often play exergames”), one participant (2.1%) rated a 1, twenty (41.7%) a 2, 24 (50.0%) a 3, two (4.2%) a 4, and one (2.1%) a 5.

The experiment took place during a gaming and cosplay event. The recruitment criteria for participating were being over the age of eighteen and attending the event with another person they knew to play as a pair. Potential participants were approached and asked if they were interested in taking part in exergame research. Interested pairs received detailed information about the study and, if willing to proceed, reviewed the materials and provided written consent. Participants were then fitted with Polar Unite fitness trackers on their wrists, filled out a questionnaire covering demographic information, and completed the international shortened version of PANAS [41]. Next, participants watched an instructional video demonstrating how to play and interact with the version of the exergame they would begin to play. Twelve pairs (24 participants) began with the cooperative mode, while the other twelve pairs started with the collaborative mode. Each pair was then positioned with the participants side by side with each other, approximately 1.5 meters from the screen, with a webcam adjusted to ensure clear visibility. Before playing, participants were asked to demonstrate the required movements and received guidance for incorrect movements until they were accurate. Each pair then played one 2-minute game round, after which the participants’ final scores were recorded. Participants then completed a questionnaire containing the PANAS scale [44], PENS questionnaire [19], Borg RPE scale [39, 40], and PACES [42, 43]. Participants were then instructed to play a second round of gameplay, this time using the other version of the exergame. After finishing the second round, participants once again filled out an identical questionnaire.

3.3 Statistical Analysis

Before running any tests, the internal consistencies of the scales were measured with Cronbach’s alpha. When analyzing the changes in emotional states after one round of the game, the positive affect scale before gameplay was slightly below the acceptable limit of 0.6 [45] at 0.56 for the group that would play the cooperative variant and 0.57 for those that would play the collaborative variant. Due to the closeness to acceptable values and high reliability after gameplay (0.77 and 0.75), we decided to move on with the analysis. For negative affect, the reliability was low for the collaborative variant group after gameplay at only 0.37 with low variance in answers. As the reliability was acceptable for the cooperative variant group both before (0.71) and after (0.63) gameplay and the collaborative variant group had high reliability before gameplay (0.68) we decided to continue with the analysis anyway. For the analysis of the difference between variants, all values were above acceptable values, with the lowest being 0.63. The Shapiro-Wilk test statistic revealed multiple violations against the assumption of

normality, which led us to use the non-parametric Wilcoxon signed rank test for the analyses.

4 Results

The collaborative group showed a significant increase in positive affect after one round of gameplay (see Table 2) with a medium to large effect [46]. Negative affect was also lower after playing which was significant with an effect size corresponding to a large effect. The participants who played the cooperative version showed similar trends with increased positive and decreased negative affect, but none of those differences were statistically significant (see Table 1). Participants in both groups had increased heart rates during gameplay compared to their resting pulses with large effect sizes [46].

The participants had higher average heart rates when playing the cooperative version compared to the collaborative (see Table 3) which was significant with a medium effect size [46]. The highest heart rate reached during the session was also significantly higher in the cooperative version with an effect size indicating a medium to large effect [46]. This was also reflected in the reported perceived exertion, as participants reported significantly higher perceived exertion in the cooperative version with an effect size indicating a medium effect [46]. The participants also performed significantly better at an individual level in the cooperative version than in the collaborative with a medium effect size [46]. While participants showed slightly higher positive affect and slightly lower negative affect in the collaborative version compared to the cooperative version, these differences were not statistically significant (see Table 3). Likewise, there were no statistically significant differences between the groups in terms of autonomy or competence. There was a difference in relatedness, where the participants reported higher levels of relatedness in the collaborative compared to the cooperative version. The effect size indicates a medium effect [46]. No difference was found between the two versions in terms of enjoyment.

4.1 Game Preferences

In the final survey, 12 participants preferred the cooperative version, 32 preferred the collaborative, and 4 stated that both versions were equally fun. Interestingly, out of the 12 participants preferring the cooperative variant, 11 started playing the collaborative version. Out of the participants preferring the collaborative version, 21 had started playing the cooperative version while 11 began with the collaborative version. When asked why the participants preferred the version they chose, participants finding the collaborative version were more focused on teamwork, and some thought the cooperative version felt competitive. This was seen in both those preferring the collaborative and those preferring the cooperative version, and the competitive aspect was seen as positive by those who enjoyed competition and negative by those preferring teamwork.

5 Discussion

Our study examined collaborative versus cooperative versions of an exergame. Previous research [15, 16] suggested positive affect would increase in both versions, but this only occurred in the collaborative version. Similarly, while we expected decreased negative affect based on prior findings [33, 41], this was only observed in the

Table 1: Collaborative version before and after the first game session

Variable	Mean (SD) Baseline	Median (IQR) Baseline	Mean (SD) After game	Median (IQR) After game	p-value (2-tailed)	z- score	Effect size r
Positive affect	14.13 (.638)	15.00 (5.00)	15.71 (.772)	15.50 (5.00)	.026 ^a	2.221	.4534
Negative affect	6.67 (.433)	6.00 (2.00)	5.46 (.170)	5.00 (1.00)	.001 ^a	3.207	.6546
Heart rate	40.65% (1.19%)	40.20% (6.56%)	55.16% (1.71%)	54.23% (13.76%)	<.001 ^a	4.286	.8749

^a Significant at $p < .05$.

Table 2: Cooperative version before and after the first game session

Variable	Mean (SD) Baseline	Median (IQR) Baseline	Mean (SD) After game	Median (IQR) After game	p-value (2-tailed)	z- score	Effect size r
Positive affect	12.83 (.622)	12.00 (6.00)	14.21 (.851)	14.00 (7.00)	.066	1.836	.3748
Negative affect	6.46 (.351)	6.00 (2.00)	6.08 (.345)	5.50 (1.00)	.319	.996	.2033
Heart rate	41.86% (1.29%)	40.92% (8.81)	57.30% (1.57%)	55.83% (11.01%)	<.001 ^a	4.286	.8749

^a Significant at $p < .05$.

Table 3: Comparison of the collaborative and cooperative exergame versions

Variable	Mean (SD) Collaborative	Median (IQR) Collaborative	Mean (SD) Cooperative	Median (IQR) Cooperative	p-value (2-tailed)	z- score	Effect size r
Heart rate (average)	56.15% (7.78%)	54.44% (10.37%)	59.16% (8.38%)	57.90% (12.48%)	.005 ^a	2.783	.4017
Heart rate (max)	65.05% (9.57 %)	63.29% (12.65%)	68.82% (10.82%)	66.92% (15.26%)	.002 ^a	3.104	.4480
Perceived exertion	11.77 (2.520)	12.00 (4.00)	12.56 (2.782)	12.50 (3.00)	.010 ^a	2.568	.3707
Performance	9.79 (4.608)	9.00 (6.00)	12.42 (6.559)	11.00 (10.00)	.005 ^a	2.824	.4076
Positive affect	15.37 (3.966)	16.00 (6.00)	15.00 (4.058)	15.00 (8.00)	.176	1.355	.1956
Negative affect	5.60 (1.216)	5.00 (1)	5.71 (1.336)	5.00 (1.00)	.653	.449	.0648
Enjoyment	29.08 (4.361)	30.00 (4.00)	28.46 (4.758)	30.00 (5.00)	.453	.750	.1082
Competence	11.67 (2.077)	12.00 (3.00)	11.50 (2.343)	12.00 (3.00)	.619	.497	.0717
Autonomy	9.31 (2.536)	10.00 (3.00)	9.06 (2.453)	9.00 (4.00)	.331	.971	.1402
Relatedness	13.10 (2.578)	13.00 (4.00)	11.92 (2.728)	11.50 (5.00)	.007 ^a	2.685	.3875

^a Significant at $p < .05$.

collaborative version. It is possible that collaborative exergames have a stronger effect on negative emotions than cooperative exergames. However, it must be noted that there were reliability issues with the negative affect scale for the collaborative group, which could affect the results. When comparing the positive and negative affect reported after each version, the collaborative version did not lead to better improvement in mood than the cooperative version.

There are a few possible explanations for why the cooperative version led to higher relative heart rates. First, the collaboration required players to stand on a button while their co-player walked through a door. For many players, this resulted in wait times where players either waited on the button for their co-player to walk through a door or waited in front of the door while waiting for their co-player to stand on a button to open it. This is not necessarily a bad thing, as some participants appreciated a brief pause in the physical activity, meaning that this game mechanic could be a way to balance the level of exertion, especially during longer play times. Second, many players perceived the cooperative

version as competitive which typically creates higher exertion [35], especially in short durations [22].

The focus on interpersonal relationships and tightly coupled teamwork [24-26] could explain the higher ratings of relatedness in the collaborative version, which aligns with previous studies showing stronger connections with higher interdependence [31]. However, no differences were found for autonomy despite cooperation being more individualistic [24-26]. Where previous studies have shown collaborative games leading to both higher challenge and enjoyment [28] [31], this study found no differences in either feelings of competence or enjoyment. The similar level of enjoyment was extra surprising considering that the collaborative version was the clear favorite for most participants. However, players also reported that they did not enjoy waiting for the other player to let them into another room. As previous work has found, higher interdependence can also become tedious and lead to tension [31, 32], potentially decreasing the overall enjoyment.

Notably, participants who preferred the cooperative version cited its competitive nature, while collaborative version supporters emphasized teamwork. In the classification by Stout and Keast

[26], cooperation is closer to competition due to their independent "power-to" structure, an aspect of cooperation also seen in other descriptions of cooperation [24, 25]. In contrast, collaboration is typically characterized by its emphasis on close teamwork in pursuit of a common goal [24-26]. These distinctions help explain why participants described the collaborative version as more teamwork oriented but also raise intriguing questions about the potential similarities between cooperative and competitive games. The findings suggest that the line between cooperation and competition in exergames may be more blurred than previously thought.

Our findings show that both versions led to an increase in heart rate, indicating some level of increased physical activity. Only the collaborative version showed a significant increase in positive and a decrease in negative affect, which might indicate that collaboration has a bigger impact on emotions. However, no difference was found in positive or negative affect in the two versions in the within-subject comparison, which, together with some issues with low internal consistency, makes it difficult to draw any strong conclusions about how collaboration and cooperation compare in terms of improving emotional states. The cooperative version led to higher levels of both objective and subjective exertion while the collaborative version led to higher feelings of relatedness. The collaborative version was preferred by most participants, and the cooperative version was not only seen as involving less teamwork but was also perceived as competitive by multiple participants, despite being informed of the objective to collect points together.

5.1 Limitations and Future Work

The study faced several limitations, including low reliability in the negative affect scale for the collaborative version, and the sample being limited to pairs who knew each other. As Chan et al. [36] pointed out, the players' personalities and preferences as well as their relationship with their co-player also affect how they experience different social interactions. By only recruiting pairs of people who already knew each other, the results might not be representative of cooperation and collaboration in pairs of strangers. Additionally, it is possible that the people who chose to participate were those who had a preference towards playing with others, thus being more inclined to enjoy collaboration [36]. This could have skewed the results towards the more collaborative version. Future research should examine the long-term effects of different teamwork types, the impact of competitiveness on teamwork experiences, the relationships between competition, cooperation, and collaboration, and the dynamics of team competitions with different teamwork styles.

Acknowledgments

The authors would like to acknowledge Tiny Jaarsma and Leonie Klompstra for their advice, feedback, and support during the writing of this paper. The authors would also like to thank everyone who participated in the study.

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